Francesco Maria Vinciguerra

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Milan, Italy & Lausanne, Switzerland

Research Experience

Visiting Researcher — AI Safety University of Warwick — Prof. **Paul Jenkins** *2026*

- Developing rare-event sampling methods for language models and masked diffusion models, with applications to discovering low-probability harmful behaviours and estimating their occurrence probabilities.
- First author of *Estimating Rare-Event Probabilities in Masked Diffusion Language Models*, accepted at the NeurIPS 2026 Trustworthy AI for Good workshop; studying the theoretical and empirical properties of the proposed estimators, including bias, variance, convergence, and computational efficiency.

Research Project in Stochastic Analysis EPFL — Prof. **Martin Hairer** *2026*

- Developed a proof strategy for a functional limit theorem for the Lorentz Mirror Walk.

Industry Experience

Data & AI Intern — AI Refinery Accenture, Milan *Jun 2025 – Sep 2025*

- Benchmarked LLMs and built evaluation pipelines to identify hallucination, reasoning, and reliability failures.
- Designed multi-agent workflows with model routing, specialized prompting, and RAG to improve grounding and task performance.
- Contributed to the AI Refinery framework by implementing agent-orchestration components and analyzing system-level performance across different configurations.

Education

EPFL — MSc in Applied Mathematics Lausanne, Switzerland *2025 – Present* **GPA: 5.2/6.**
Relevant coursework: *Stochastic Simulation, Statistical Inference, Deep Learning, Large Deviations, Optimal Transport.*

Bocconi University — BSc in Mathematical and Computing Sciences for AI Milan, Italy *2022 – 2025* **GPA: 28/30.** Thesis: *Geometric Group Theory*.

Publications

Francesco Maria Vinciguerra, Nicola Branchini. *Estimating Rare-Event Probabilities in Masked Diffusion Language Models*. Accepted at the **Trustworthy AI for Good Workshop, NeurIPS 2026** (Poster). [OpenReview]

Technical Skills & Awards

Programming & Tools: Python, PyTorch, NumPy/SciPy, Git.

Awards: IMC Trading Challenge — 1st Place (EPFL Forum 2025 Edition).

Languages: Italian (Native), English (C1).